higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – BOILER MAKING
NQF LEVEL 3

(6031013)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 9 pages and 1 diagram sheet.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 FIGURE 1 indicates a welding joint (Oblique view) with specific weld information.

Draw the Front view of the joint and add the welding symbol indicating all weld symbols on the welding symbol as indicated.

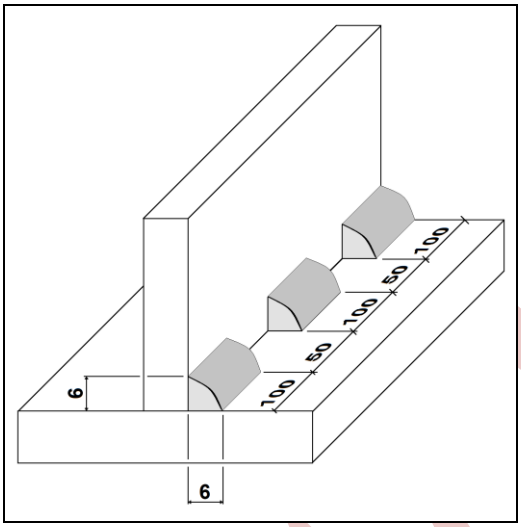


FIGURE 1

(6 × 1) (6)

1.2 FIGURES 2.1 and 2.2 indicate certain marking-off tools.

Name these tools and explain ONE use of each. Write the FIGURE number, the name of the tool and use next to the question number in the ANSWER BOOK.

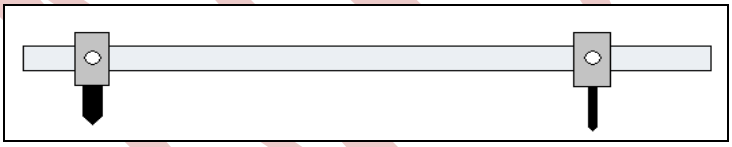


FIGURE 2.1

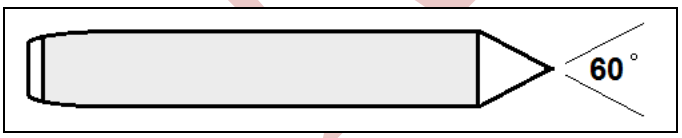
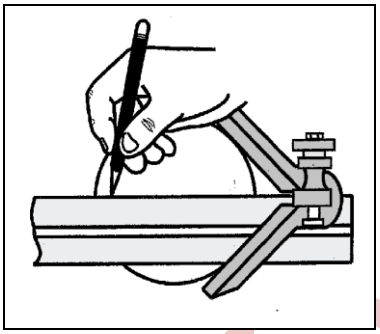


FIGURE 2.2

(2 × 2) (4)

1.3 FIGURE 3 indicate a marking-off process.

Name the tool used and explain the task being performed. (2 × 1) (2)



[Adapted from: *Eng. Fab L3 Pearson,* 2008]

FIGURE 3

1.4 Before erecting a structure, you have to match and compare the structure components to the drawing to ensure the accuracy of the structure.

Explain FOUR points that can be checked. (4 × 1) (4)

1.5 Welding procedure specifications will be developed by the welding engineers before a welding process can commence. This document provides detailed information on the welding conditions or needs.

List FOUR instructions that the welding procedure will describe and what it will be in the welding procedure. (4 × 1) (4)

[20]

QUESTION 2

2.1 When using tools ensure that they are I good condition in order to protect you from injury.

List THREE golden rules when using tools. (3 × 1) (3)

2.2 GMAW has a few disadvantages.

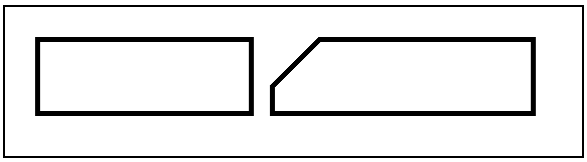
List FOUR disadvantages of gas metal arc welding. (4 × 1) (4)

2.3 Caring and storing of welding equipment are very important because damages can occur unexpectedly.

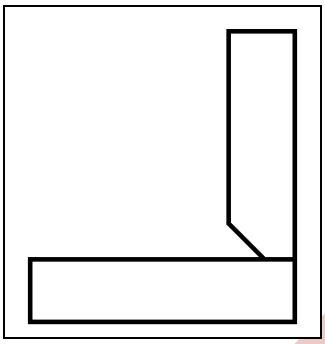
Name TWO procedures that must be followed when caring and storing of welding metal. (2 × 1) (2)

2.4 Specify the following welding joints:

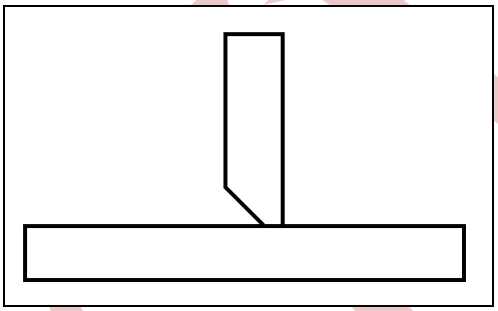
2.4.1



2.4.2



2.4.3



(3 × 2) (6)

2.5 Draw a clear and neat isometric drawing of a butt joint between two plates, indicating a few tack welds on the joint. (3 × 1) (3)

2.6 During a welding process, some defects can occur, like undercuts. Spatter is also one of the defects.

Explain THREE ways spatter can be prevented. (3 × 1) (3)

2.7 Explain in short what is meant by the defect term *porosity*. (2)

2.8 It is a reality that electrical dangers play an important role in a welding work shop. All electrical devices are dangerous especially high voltage machines. Certain general checks and preventions must be done or taken regularly.

Explain FOUR electrical checks or preventions that must be done or taken regularly. (4 × 1) (4)

2.9 Alloys are mixtures of two or more metals.

Name the TWO metals the alloy brass is made up from. (2 × 1) (2)

2.10 Name one example of a *non-ferrous* metal. (1)

[30]

QUESTION 3

3.1 FIGURE 4 illustrate a rectangular pressure tank. It is sealed with a top lid and bottom base. The plate thickness is 20 mm.

Calculate the total area of the plate needed to manufacture the pressure tank.

NOTE: The dimensions represent the outside sizes of the tank and being given in meter units. Ensure to take the joints in consideration. Make use of the labelled surfaces to ease the task. No marks will be allocated for sub-calculations. Give the answer in m^2.

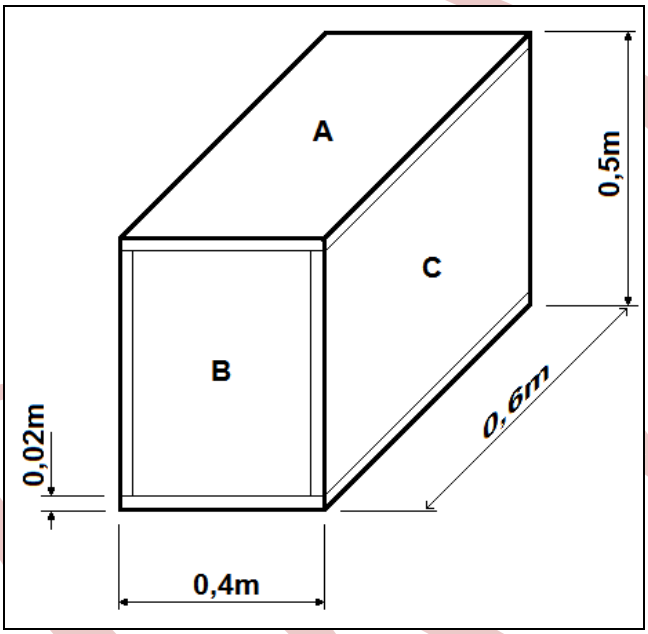


FIGURE 4

(4 × 3) (12)

3.2 FIGURE 5 illustrate an opening of a round pipe.

Construct a similar FIGURE and indicate the *mean circumference* and *mean diameter* of the pipe on the sketch.

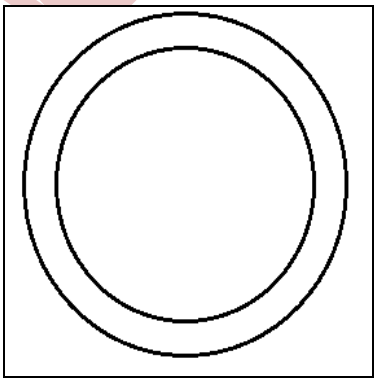


FIGURE 5

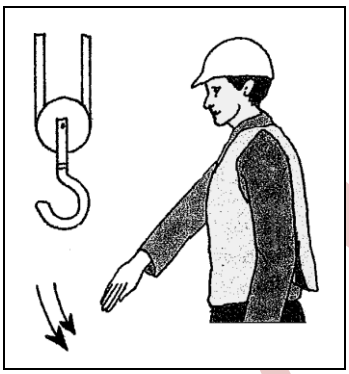
(3 × 1) (3)

3.3 During Rigging work, crane drivers can make use of high technological equipment like Remote Control Radios or Infra-red Control to assist them with the execution of their tasks.

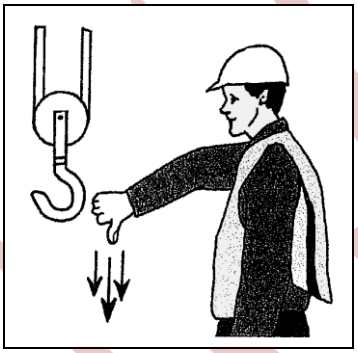
State THREE advantages of using remote or infra-red control when operating and overhead crane from the workshop floor. (3 × 1) (3)

3.4 Study the following FIGURES indicating specific hand signals used by crane driver assistants. Identify each signal.

3.4.1



3.4.2



(2 × 1) (2)

3.5 State TWO types of forming machines used for rolling cylinders in industry. (2 × 1) (2)

3.6 OH&SA ensure that employers and employees adhere to safety regulations at all times when working.

List THREE purposes of the OH&SA. (3 × 1) (3)

[25]

QUESTION 4

4.1 When manufacturing structures, you are allowed to make use of tolerances, due to the fact of human errors.

Briefly explain the meaning of the term *tolerance*. (2)

4.2 Allowances are needed for specific reasons. Without an allowance, some bending tasks cannot be done. Explain what allowance refers to. (2)

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4.3 FIGURE 6 indicate the Front view and Top view of a Right cone. Study the drawings and identify the SIX true length lines that can be observed directly from both the views.

Copy the given table, FIGURE 6.1 into your ANSWER BOOK and label the true length lines in the correct column. e.g.:

FRONT VIEW
OA

(6 × 1) (6)

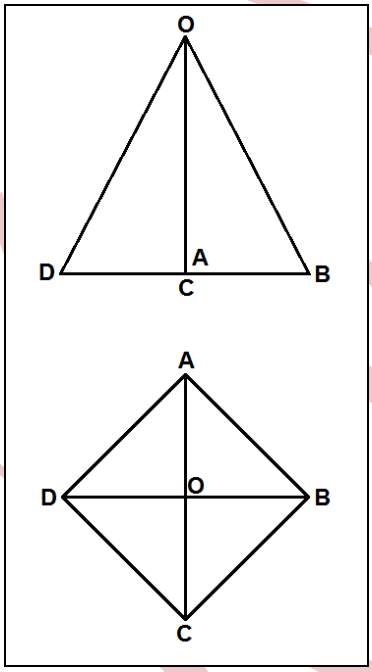


FIGURE 6

FRONT VIEW	TOP VIEW

FIGURE 6.1

4.4 The FIGURE on the attached DIAGRAM SHEET, illustrates an oval to round chute. The bending lines are partially indicated.

Make use of this drawing to construct the true lengths of bending lines 1B and 3D. The indicated reference lines must be used to determine the true lengths.

NOTE: After completing the drawing, remove the Diagram Sheet and attach it to your answer book. Ensure candidate information.

(4 × 1) (4)

4.5 Name any TWO components of the bar folding machine (*Pan Box* folding machine). (2 × 1) (2)

4.6 Explain TWO safety precautions to be taken in account when planning to use the bending machine (press brake machine). (2 × 1) (2)

4.7 Name the process that is used to remove any free iron contamination of stainless steel surface or applying a corrosion resistance film to a surface that must be welded. (1)

4.8 List FOUR welding defects that can be classified as a welding irregularity. (4 × 1) (4)

4.9 Briefly explain the reason why stainless steel have to go through pickling and passivation process. (2 × 1) (2)

[25]

TOTAL: 100

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DIAGRAM SHEET 1

QUESTION 4.4

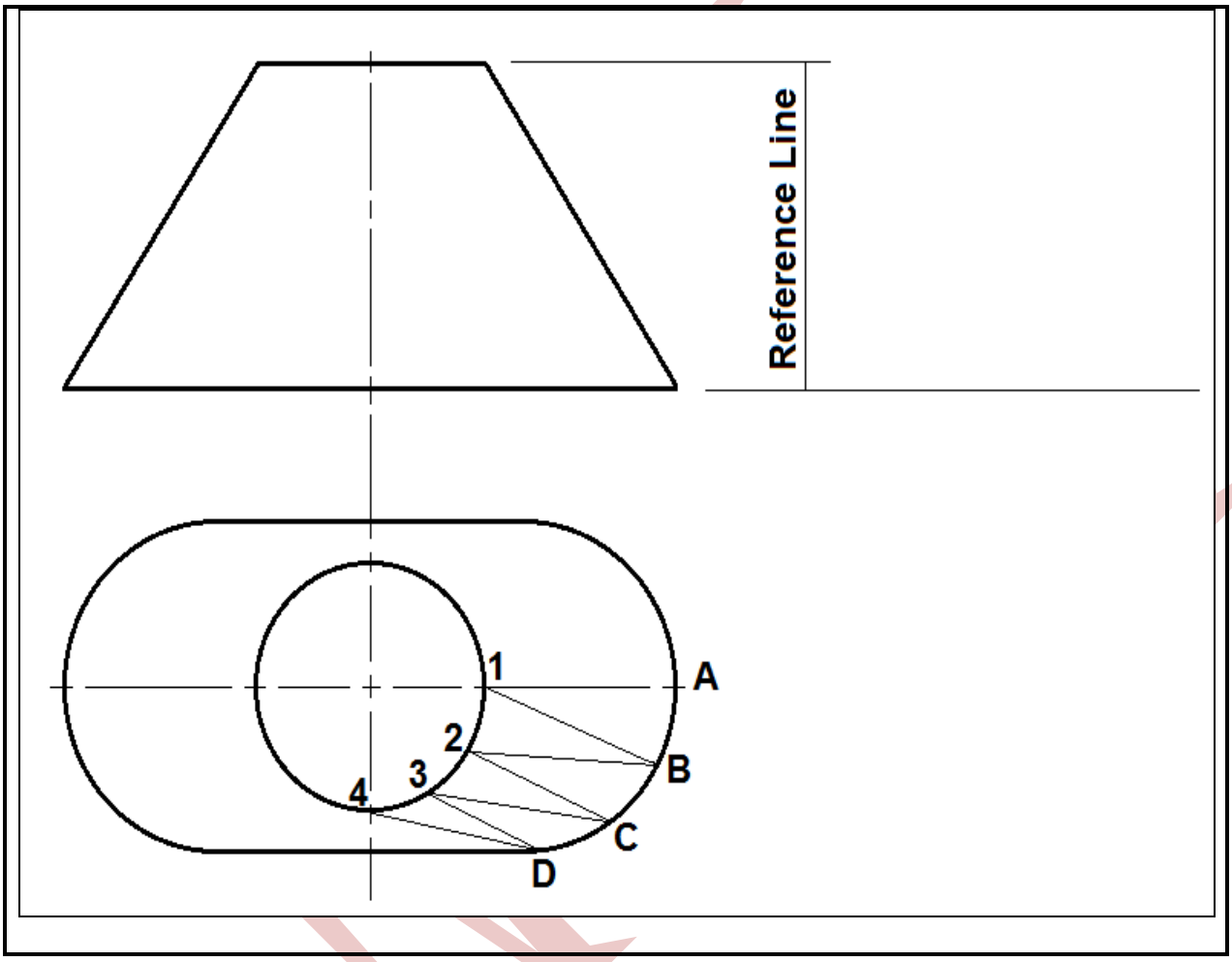


FIGURE 7

NOTE: After completing the drawing, remove this DIAGRAM SHEET and attach it to your ANSWER BOOK. Ensure candidate information.

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